Xingtian (James) SI

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EDUCATION

University of Chicago, Kenneth C. Griffin Department of Economics **Chicago, IL**

- M.A. in Economics 09/2026 – 06/2027

University of Nottingham Ningbo China, School of Humanities and Social Sciences **Ningbo, CHN**

- Bachelor of Science in Economics, First Class Honours (Top 5%) 09/2022 – 07/2026
- Final mark: 79/100; WES-evaluated GPA: 3.90/4.00; dissertation: 84/100.
- Awards: UNNC Provost's Scholarship for Academic Excellence (Top 5%, 2023-2025), Zhejiang Provincial Scholarship (2025), 2026 UNNC Award for Outstanding Students (UG), 2026 Zhejiang Excellent Graduates

University of Nottingham, Faculty of Social Science **Nottingham, UK**

- Economics (Exchange Program) GPA: 4.0/4.0 09/2024 – 01/2025

PUBLICATIONS

Si, Xingtian (2026). "Causal Effects of Coupon Expiration on Consumer Acceptance: Evidence from Heterogeneous Treatment Effects in an Observational Scenario Survey." ***Applied Economics***, *pp. 1–20.*

Si, Xingtian (2025). Estimating Heterogeneous Treatment Effects of Early Childhood Interventions on Fifth-Grade Math Achievement: A Machine Learning-Augmented Causal Analysis of ECLS-K:2011. ***International Journal of Intelligence Science***, 15, 145–161.

Si, Xingtian., *Lin, Keren.*, *Yang, Xuye.*, *Wen, Zihan.* (2025). Board Gender Diversity on Firm Performance in the UK: Evidence of Non-linear Returns and Firm Size Moderation (2008–2016). ***Entrepreneurship and Innovation***. 2025, 09(04): 141-145.

WORKING PAPER

Si, Xingtian., Fang, Chunxi., Xie, Wenhui. "When Click-to-Cancel Creates Complexity: A Theory of Subscription Regulation." Working paper, June 2026.

Si, Xingtian. "When Does Plug-in Data Thinning Work? Validity Boundaries, Misspecification and Choose-then-Infer in Econometric Panels." Working paper, October 2026.

RESEARCH EXPERIENCE

Correspondence Audit Studies on Labor Market Discrimination **San Diego, CA**

Research Assistant to Prof. Sally Sadoff, Rady School of Management, UC San Diego. *05/2026 – Present*

- Systematically reviewed 22 correspondence audit studies to document how fictitious names signal race and gender in labor-market discrimination research; constructed and audited a 53-sheet database of 600+ names, including name pairings and source metadata.
- Harmonized eight name-perception studies into a dataset of 1,600+ names and 25 variables, including race and gender, perception accuracy and birth-record frequencies; prepared a data dictionary and cross-study overlap analysis.
- Processed data from 10,000+ respondents and 44,000+ observations in Python to construct race and gender perception accuracy measures and an analysis-ready participant-level dataset.

When Click-to-Cancel Creates Complexity: A Theory of Subscription Regulation **Ningbo, CHN & Chicago, IL**

Co-authored Research, supervised by Assoc. prof. Haibo Xu of the University of Nottingham Ningbo China *04/2026 – Present*

- Developed a theoretical model in which firms jointly choose cancellation friction and retention-menu complexity. Derived a substitution mechanism: tighter cancellation restrictions expose more consumers to retention offers, increasing the return to complexity in the baseline model.
- Derived conditions under which induced complexity offsets the consumer benefits of easier cancellation, incorporating time costs borne by consumers who ultimately cancel. Constructed a solved example showing that the consumer-welfare-maximizing friction cap can be interior.
- Extended the model to examine friction-induced fatigue and joint regulation of both instruments. Characterized when the complexity response can reverse and showed how joint constraints eliminate endogenous complexity adjustment while leaving consumers exposed to existing complexity.

When Does Plug-in Data Thinning Work? **Chicago, IL**

Independent Research *08/2026 – Present*

- Derived an asymptotic characterization of the dependence induced by plug-in covariance estimation in clustered panels, demonstrating that consistent covariance estimation does not guarantee joint independence of thinned data components.
- Established sufficient conditions for asymptotically valid post-selection inference based on fixed-dimensional linear aggregates, and characterized how aggregation, covariance misspecification, and error-direction alignment affect inferential validity.
- Conducted Monte Carlo experiments involving difference-in-differences, event-study pretesting, and covariate screening, alongside a county-level minimum-wage placebo application, to evaluate conditional rejection rates and compare plug-in thinning with cluster splitting, exact rotation, and conditional inference.

Causal Machine Learning and Counterfactual Inference in Consumer Behavior **Ningbo, CHN**

Dissertation, supervised by Assoc. Prof. Zhenjiang Lin of the University of Nottingham Ningbo China *09/2025 – 05/2026*

- Studied two-hour versus one-day coupon windows using 12,007 scenario-survey responses across 490 demographic profiles. Under selection on observables, cross-fitted IPW estimates an 18.3 percentage point reduction in stated acceptance (cluster-robust SE: 1.4 pp).
- Separated heterogeneity discovery from confirmation through profile-cluster sample splitting, using causal forests and T-learners to explore effect variation and a held-out sample to evaluate contrasts by urgency, social context and distance.
- Compared effect-based targeting with propensity-based and random allocation under a fixed short-expiration coupon budget; simulations showed 29–42% lower acceptance losses than random allocation, with uncertainty assessed using 500-replication cluster-bootstrap intervals.

Early Childhood Intervention and Mathematics Achievement **New York, NY**

Research Assistant to Dr. Jiaxi Yang (Ph.D., Columbia University) *03/2025 – 09/2025*

- Analyzed baseline differences between special-education and general-education cohorts across six demographic, academic, and family-related domains, using standardized mean differences and variance ratios to assess group comparability and potential selection bias.
- Applied causal random forests to investigate heterogeneous effects of special-education services on fifth-grade mathematics achievement, examining variation in estimated treatment effects across students with different socioeconomic and academic backgrounds.
- Developed interactive dashboards in R Shiny and Python Dash to visualize subgroup-level treatment-effect estimates and support the interpretation of potential policy implications.

Do Female Directors Improve the Financial Performance of Companies **Nottingham, UK**

Team Member, supervised by Prof. Abigail Barr, University of Nottingham *11/2024 – 06/2025*

- Conducted econometric analysis of 840 UK firm-year observations (2008-2016) using Stata to examine the relationship between board gender diversity and corporate financial performance, measured by ROA and Tobin's Q.
- Compared linear and nonlinear regression specifications to investigate potential nonlinear associations, finding an inverted-U relationship between female board representation and Tobin's Q, with the estimated turning point around 20-25%.

INTERNSHIP EXPERIENCE

Shenzhen Capital Group (SCGC) **Hangzhou, CHN**

Assistant Investment Manager in Southeast Investment Department. *07/2025 – 09/2025*

- Prepared 10+ industry research and investment materials and analyzed 10+ start-ups to support investment screening.

Guotai Junan Securities Co., Ltd **Shanghai, CHN**

Investment Banking Analyst in Investment Banking Department *07/2024 – 08/2024*

- Contributed to 5+ prospectuses and listing documents, financial due diligence, and analysis of 10+ rejected listing cases.

HONORS

Honorable Mention, Mathematical Contest In Modeling (2026)
Second Prize, “Huashu Cup” International Mathematical Modeling Competition (2026)
Second Prize, National College Student Mathematical Modeling Competition (2025)
Second Prize, “Huashu Cup” National Mathematical Modeling Competition (2025)

SKILLS

Programming: Python, R, Stata, SQL, MATLAB, LaTeX